

Set 11: Product Category Analysis

- a) Generate a funnel chart to analyze the sales conversion process for each product category. Label the stages and title the chart.

R code: # Install packages if not installed

```
install.packages("plotly")
```

```
# Load library
```

```
library(plotly)
```

```
# Funnel Chart
```

```
plot_ly(
```

```
  type = "funnel",
```

```
  y = product_data$Category,
```

```
  x = product_data$Sales,
```

```
  textinfo = "value+percent initial"
```

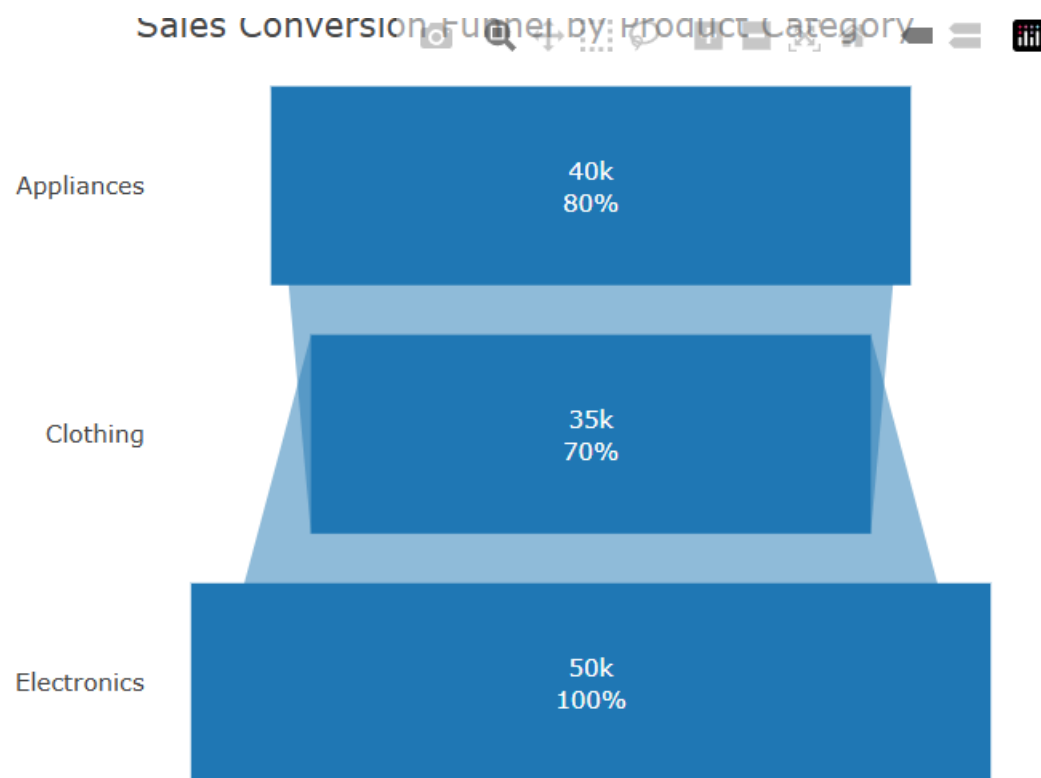
```
) %>%
```

```
layout(
```

```
  title = "Sales Conversion Funnel by Product Category"
```

```
)
```

Output:



- b) Build a table to display the sales data for each product category. Label the table elements.

R code:

```
# Display the table
```

```
print(product_data)
```

```
# Optional: Better formatted table
```

```
install.packages("gridExtra")
```

```
library(gridExtra)
```

```
grid.table(product_data)
```

Output:

	Category	Sales
1	Electronics	50000
2	Clothing	35000
3	Appliances	40000

- c) Develop a Tableau dashboard combining the pie chart, funnel chart, and the table for interactive exploration of product category data.

R code:

Install packages

```
install.packages(c("plotly","gridExtra"))
```

```
library(plotly)
```

```
library(gridExtra)
```

Pie Chart

```
pie_chart <- plot_ly(
  labels = product_data$Category,
  values = product_data$Sales,
  type = "pie",
  textinfo = "label+percent"
)
```

Funnel Chart

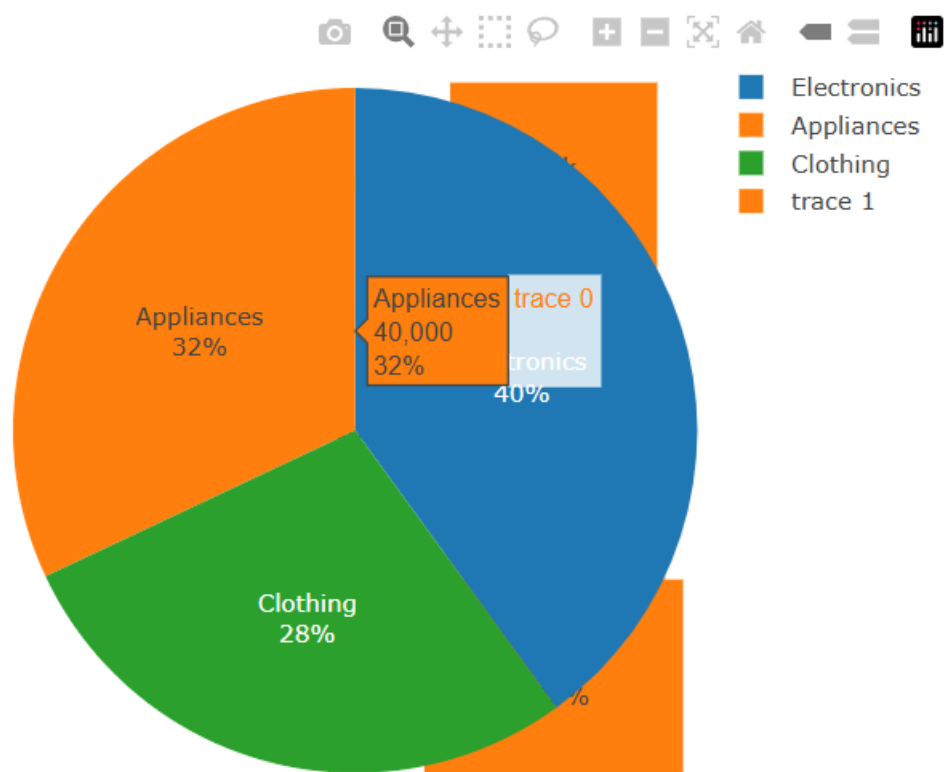
```
funnel_chart <- plot_ly(
  type = "funnel",
  y = product_data$Category,
  x = product_data$Sales,
  textinfo = "value+percent initial"
```

)

```
# Display both charts together
```

```
subplot(  
  pie_chart,  
  funnel_chart,  
  nrows = 1,  
  margin = 0.08,  
  titleX = TRUE,  
  titleY = TRUE  
)
```

Output:



- d) Create a pie chart to represent the distribution of sales across product categories. Include labels.

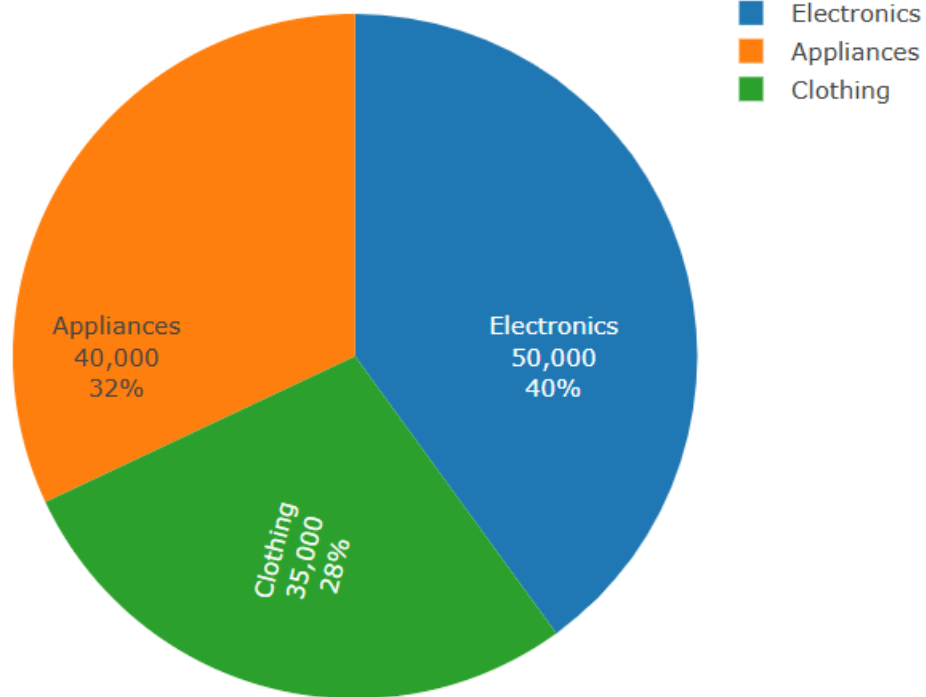
R code:

```
# Load library
library(plotly)

plot_ly(
  data = product_data,
  labels = ~Category,
  values = ~Sales,
  type = "pie",
  textinfo = "label+percent+value",
  insidetextorientation = "radial"
) %>%
layout(
  title = "Sales Distribution Across Product Categories"
)
```

Output:

Sales Distribution Across Product Categories



Set 12:Dataset: Website Traffic

- Create a line chart to visualize the trend in daily page views over time. Label the axes and title the chart.

R code:

```
# Load library
```

```
library(ggplot2)
```

```
# Line Chart
```

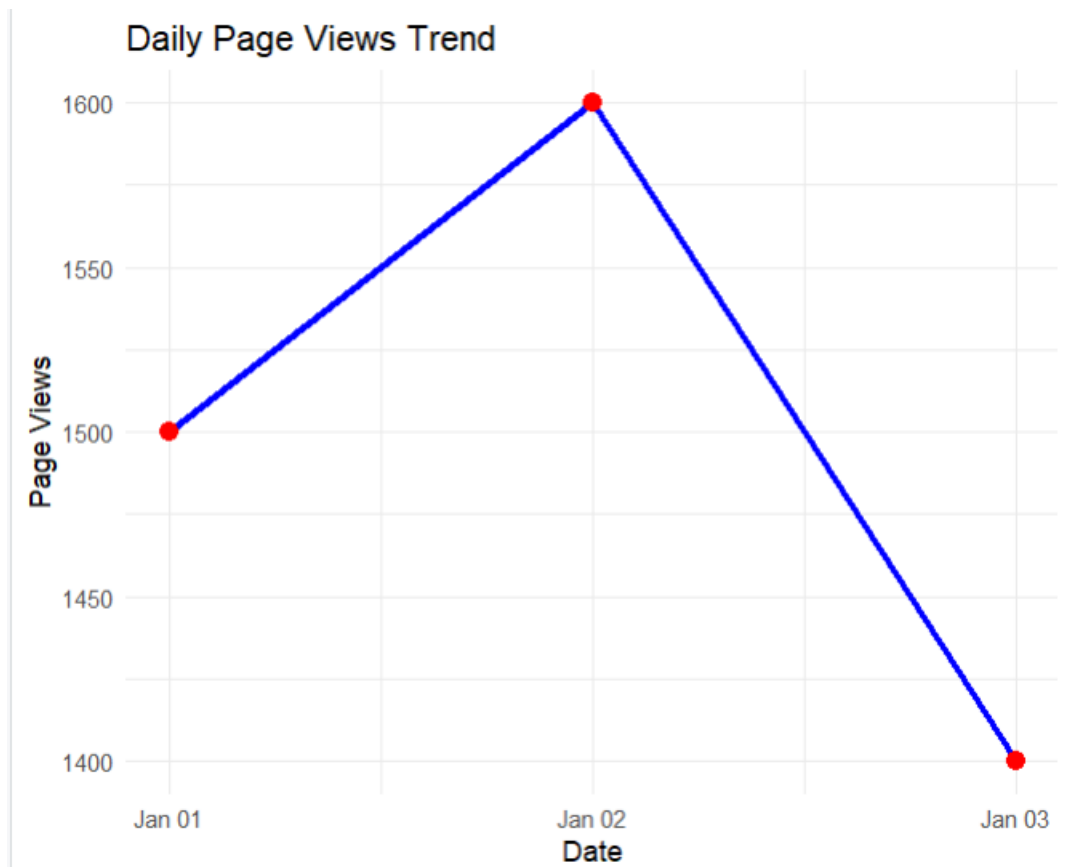
```
ggplot(traffic_data,  
  aes(x = Date, y = Page_Views, group = 1)) +  
  geom_line(color = "blue", linewidth = 1.2) +  
  geom_point(color = "red", size = 3) +  
  labs(  
    title = "Daily Page Views Trend",
```

```

x = "Date",
y = "Page Views"
) +
theme_minimal()

```

Output:



- b) Generate a bar chart showing the top N days with the highest click-through rates. Label the chart elements.

R code: `# Sort data by CTR (Highest first)`

```
top_ctr <- traffic_data[order(-traffic_data$CTR), ]
```

`# Bar Chart`

```
ggplot(top_ctr,
```

```
  aes(x = reorder(as.character(Date), CTR), y = CTR)) +
```

```
  geom_bar(stat = "identity", fill = "darkgreen") +
```

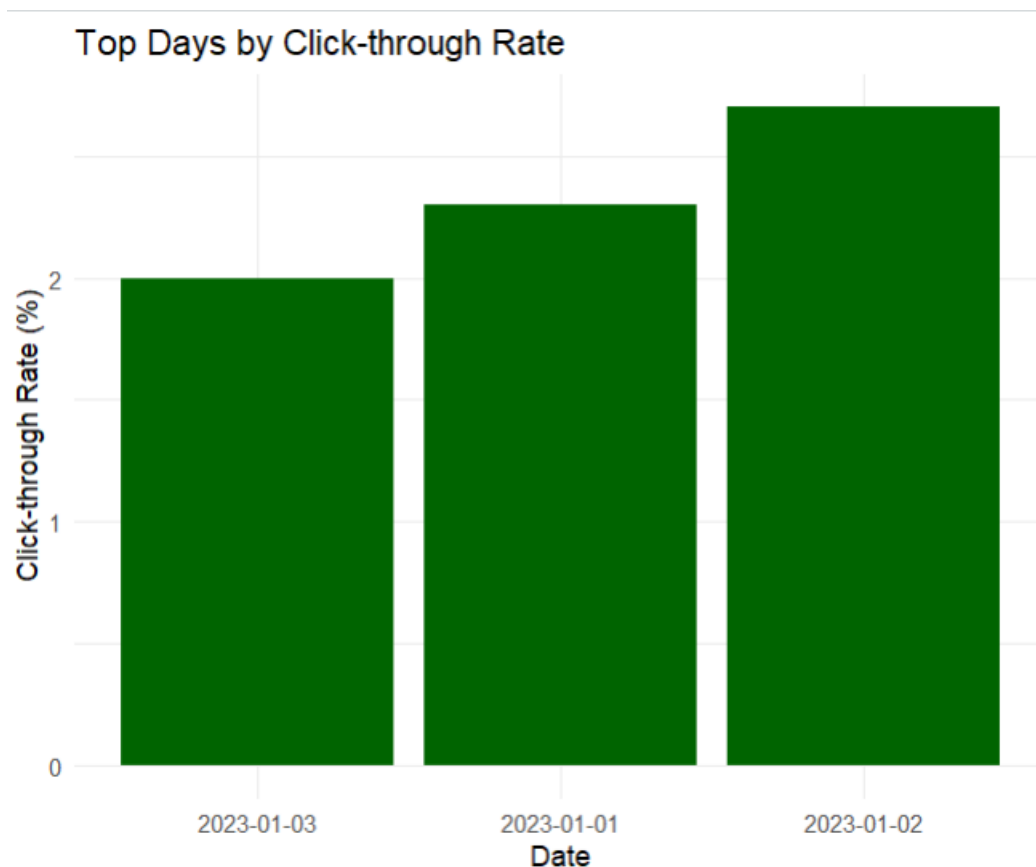
```
  labs(
```

```

title = "Top Days by Click-through Rate",
x = "Date",
y = "Click-through Rate (%)"
) +
theme_minimal()

```

Output:



- c) Build a table to show the website traffic data for each day. Label the table elements.

R code:

```
# Display dataset
```

```
print(traffic_data)
```

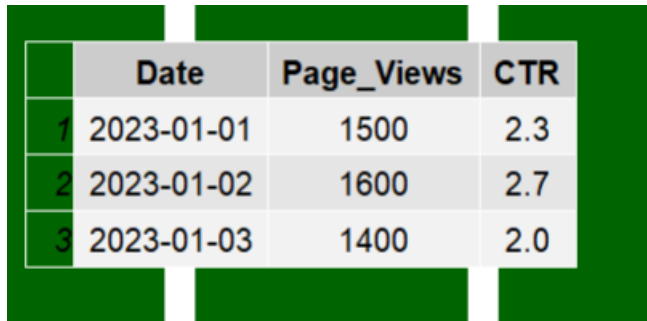
```
# Optional formatted table
```

```
install.packages("gridExtra")
```

```
library(gridExtra)
```

```
grid.table(traffic_data)
```

output:



	Date	Page_Views	CTR
1	2023-01-01	1500	2.3
2	2023-01-02	1600	2.7
3	2023-01-03	1400	2.0

- d) Develop a Tableau dashboard combining the line chart, bar chart, and the table for interactive exploration of website traffic data.

R code:

```
# Install packages
```

```
install.packages(c("plotly", "gridExtra"))
```

```
# Load libraries
```

```
library(plotly)
```

```
library(gridExtra)
```

```
# Line Chart
```

```
line_chart <- plot_ly(
```

```
  data = traffic_data,
```

```
  x = ~Date,
```

```
  y = ~Page_Views,
```

```
  type = "scatter",
```

```
  mode = "lines+markers",
```

```
  line = list(color = "blue")
```

```
) %>%
```

```
layout(
```

```

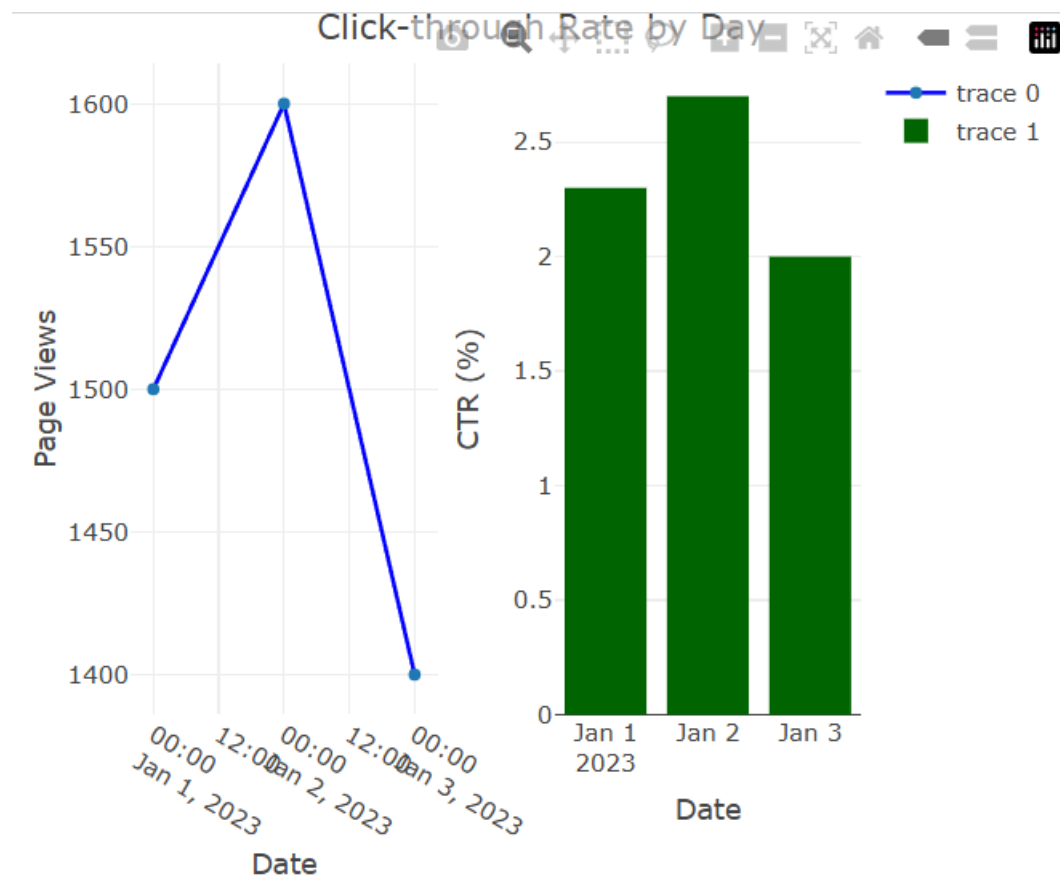
    title = "Daily Page Views Trend",
    xaxis = list(title = "Date"),
    yaxis = list(title = "Page Views")
)

# Bar Chart
bar_chart <- plot_ly(
  data = traffic_data,
  x = ~Date,
  y = ~CTR,
  type = "bar",
  marker = list(color = "darkgreen")
) %>%
layout(
  title = "Click-through Rate by Day",
  xaxis = list(title = "Date"),
  yaxis = list(title = "CTR (%)")
)

# Dashboard
subplot(
  line_chart,
  bar_chart,
  nrows = 1,
  margin = 0.08,
  titleX = TRUE,
  titleY = TRUE
)

```

Output:



Set 13: Geographic Data

- a) Create a map chart to visualize the distribution of cities on a geographic map.
Label the map elements.

R code:

```
# Install packages if not installed
```

```
install.packages(c("leaflet"))
```

```
# Load library
```

```
library(leaflet)
```

```
# Create Map
```

```
leaflet(geo_data) %>%
```

```
  addTiles() %>%
```

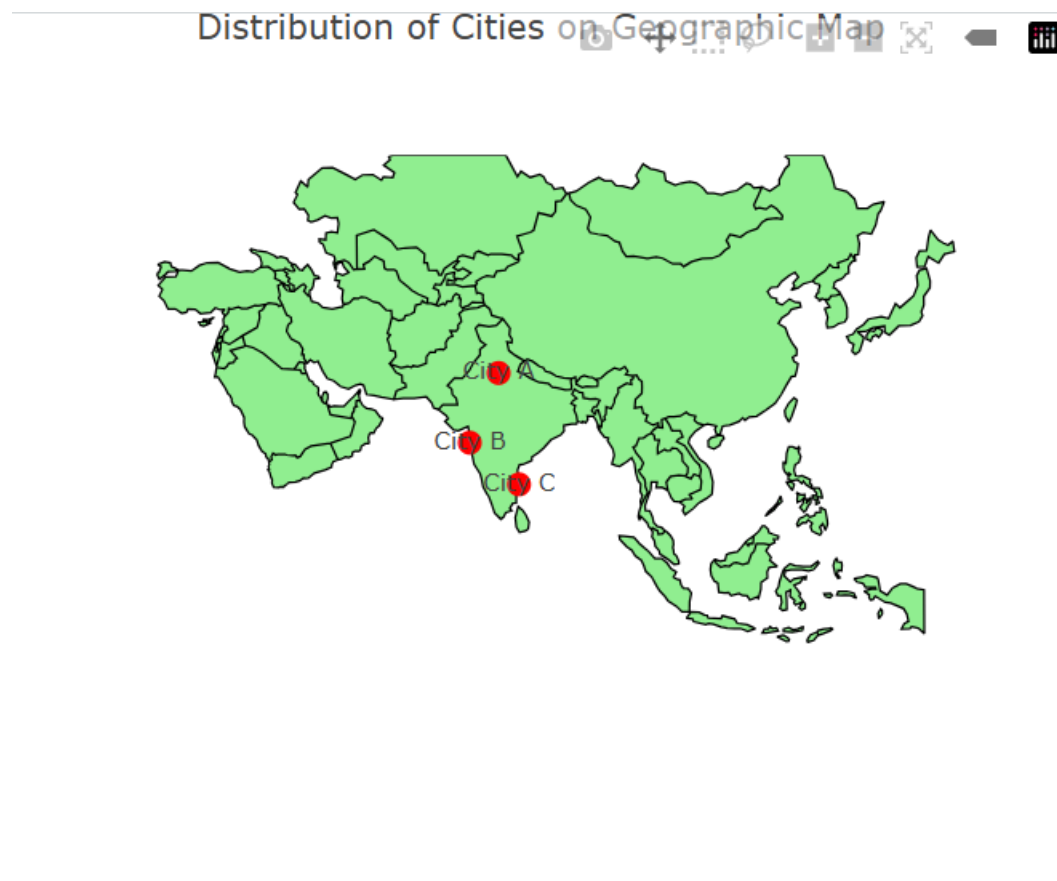
```
  addMarkers(
```

```

lng = ~Longitude,
lat = ~Latitude,
popup = ~paste(
  "<b>", City, "</b><br>",
  "Population: ", Population, "<br>",
  "Temperature: ", Avg_Temperature, "°F<br>",
  "Elevation: ", Elevation, " m"
)
)
)

```

Output:



- b) Generate a scatter plot to explore the relationship between average temperature and population size. Explain any insights.

R code:

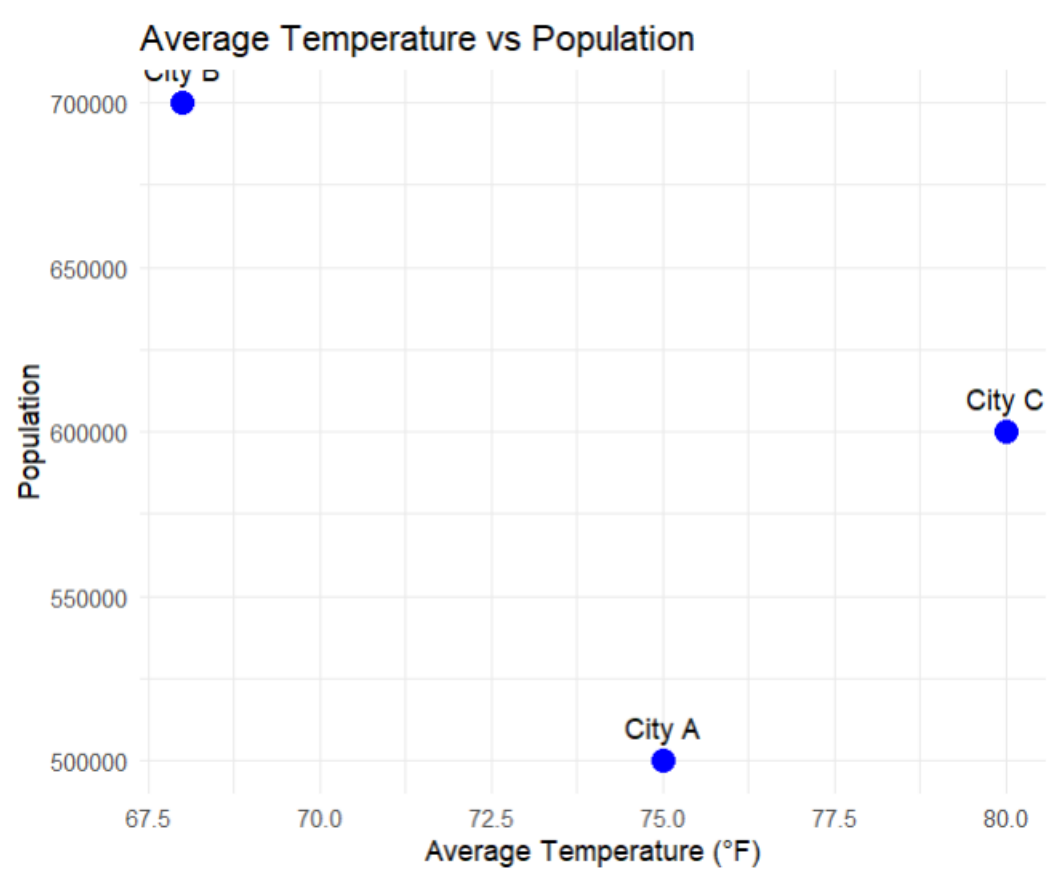
```

# Install package if not installed
install.packages("ggplot2")

```

```
# Load library
library(ggplot2)

ggplot(geo_data,
       aes(x = Avg_Temperature,
           y = Population,
           label = City)) +
  geom_point(size = 4, color = "blue") +
  geom_text(vjust = -1) +
  labs(
    title = "Average Temperature vs Population",
    x = "Average Temperature (°F)",
    y = "Population"
  ) +
  theme_minimal()
output:
```



- c) Build a table to display the geographic data for each city. Label the table elements.

R code:

```
# Display dataset
```

```
print(geo_data)
```

```
# Optional formatted table
```

```
install.packages("gridExtra")
```

```
library(gridExtra)
```

```
grid.table(geo_data)
```

output:

City	Population	Avg_Temperature	Elevation	Latitude	Longitude
City A	5e+05	75	1000	28.6139	77.209
City B	7e+05	68	800	19.0760	72.877
City C	6e+05	80	1200	13.0827	80.270

- d) Develop a Tableau dashboard combining the map chart, scatter plot, and the table for interactive exploration of geographic data.

R code:

Install packages

```
install.packages(c("plotly", "gridExtra"))
```

Load libraries

```
library(plotly)
```

```
library(gridExtra)
```

Scatter Plot

```
scatter_plot <- plot_ly(
```

```
  data = geo_data,
```

```
  x = ~Avg_Temperature,
```

```
  y = ~Population,
```

```
  type = "scatter",
```

```
  mode = "markers+text",
```

```
  text = ~City,
```

```
  textposition = "top center",
```

```
  marker = list(size = 12, color = "blue")
```

```
) %>%
```

```
layout(
```

```
  title = "Temperature vs Population",
```

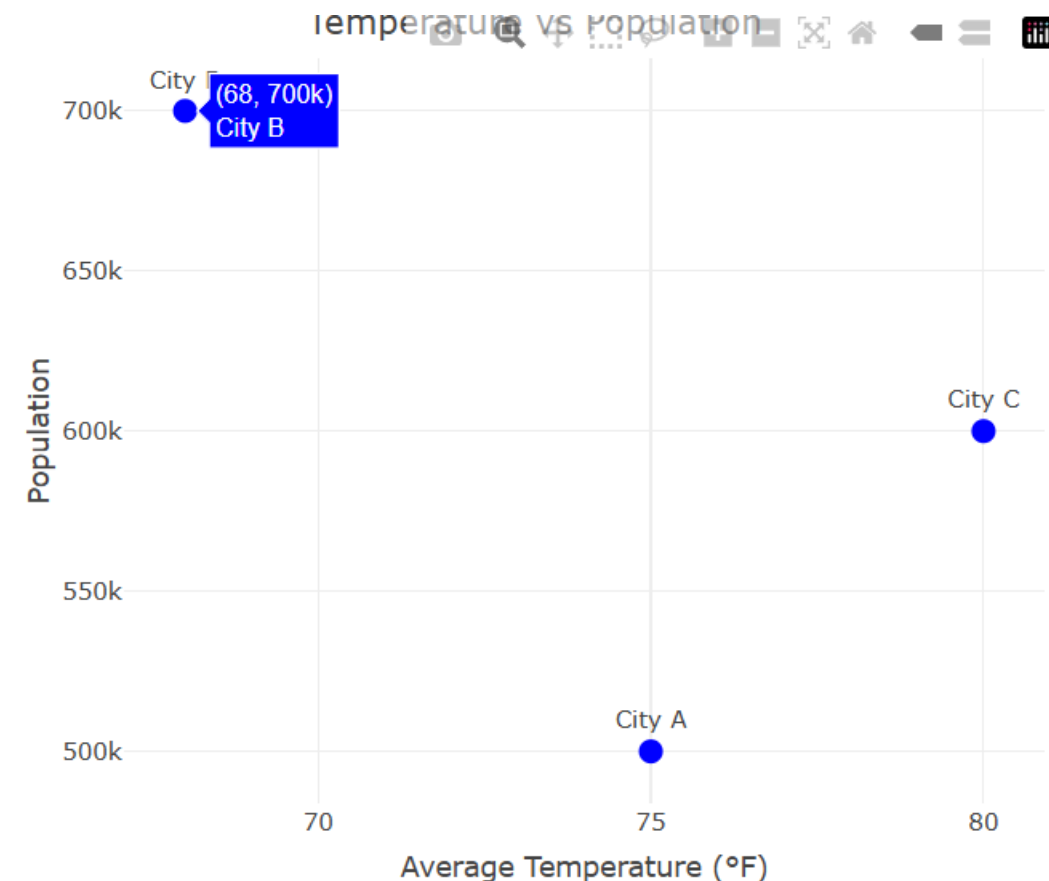
```
  xaxis = list(title = "Average Temperature (°F)",
```

```
yaxis = list(title = "Population")
)
```

```
# Display Scatter Plot
```

```
scatter_plot
```

```
output:
```



Set 14: Survey Results

- Create a stacked bar chart to display the distribution of answers for Question 1. Label the chart elements.

R code:

```
# Install packages if not installed
```

```
install.packages(c("ggplot2", "dplyr"))
```

```
# Load libraries
```

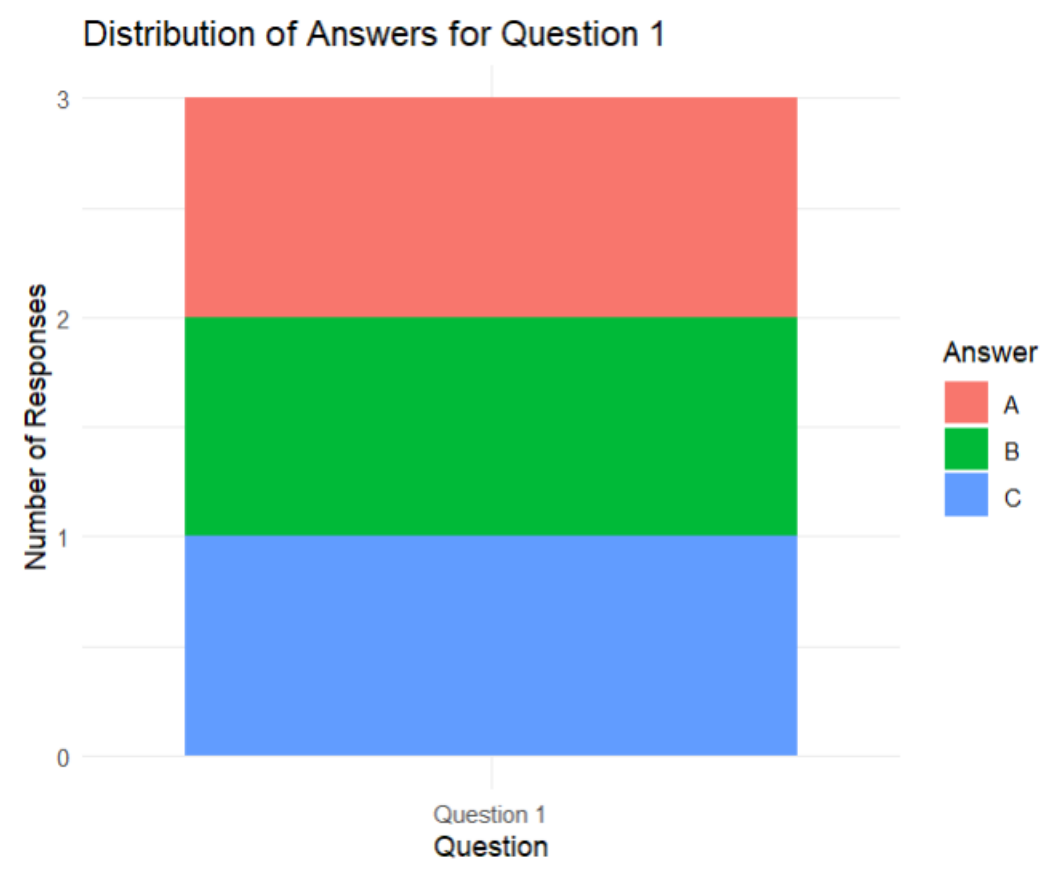
```
library(ggplot2)
```

```
library(dplyr)

# Count responses for Question 1
q1_data <- survey_data %>%
  count(Question1)

# Stacked Bar Chart
ggplot(q1_data,
       aes(x = "Question 1", y = n, fill = Question1)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Question",
    y = "Number of Responses",
    fill = "Answer"
  ) +
  theme_minimal()
```

Output:



- b) Generate a radar chart to visualize the overall pattern of responses across all three questions.

R code:

```
# Install packages
```

```
install.packages("fmsb")
```

```
# Load library
```

```
library(fmsb)
```

```
# Convert responses into numeric values
```

```
response_map <- c(A = 1, B = 2, C = 3, D = 4)
```

```
radar_data <- data.frame(
```

```
  Q1 = response_map[survey_data$Question1],
```

```
  Q2 = response_map[survey_data$Question2],
```

```
Q3 = response_map[survey_data$Question3]  
)
```

```
# Add maximum and minimum values
```

```
radar_chart <- rbind(  
  c(4,4,4),  
  c(1,1,1),  
  radar_data  
)
```

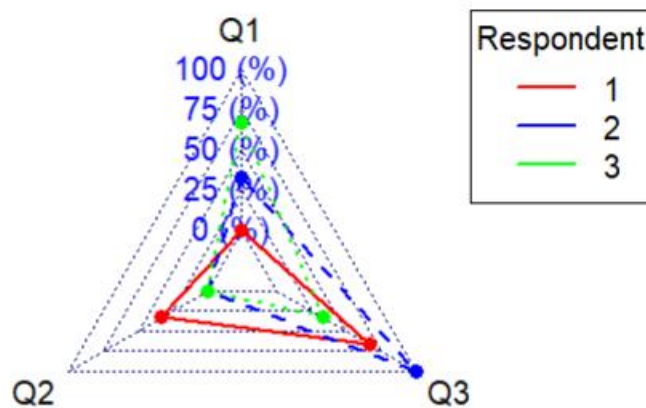
```
# Radar Chart
```

```
radarchart(  
  radar_chart,  
  axistype = 1,  
  pcol = c("red","blue","green"),  
  plwd = 2,  
  title = "Survey Response Pattern"  
)
```

```
legend(  
  "topright",  
  legend = survey_data$Respondent,  
  col = c("red","blue","green"),  
  lty = 1,  
  lwd = 2,  
  title = "Respondent"  
)
```

Output:

Survey Response Pattern



- c) Build a table to show the survey response data for each respondent. Label the table elements.

R code:

```
# Display dataset
```

```
print(survey_data)
```

```
# Optional formatted table
```

```
install.packages("gridExtra")
```

```
library(gridExtra)
```

```
grid.table(survey_data)
```

output:

	Respondent	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

- d) Develop a Tableau dashboard combining the stacked bar chart, radar chart, and the table for interactive exploration of survey responses.

R code:

Install packages

```
install.packages(c("plotly", "gridExtra"))
```

Load libraries

```
library(plotly)
```

```
library(gridExtra)
```

Count Question 1 responses

```
q1_data <- as.data.frame(table(survey_data$Question1))
```

```
colnames(q1_data) <- c("Answer", "Count")
```

Interactive Bar Chart

```
bar_chart <- plot_ly(
```

```
  q1_data,
```

```
  x = ~Answer,
```

```
  y = ~Count,
```

```
  type = "bar",
```

```
  marker = list(color = "steelblue")
```

```
) %>%
```

```
layout(
```

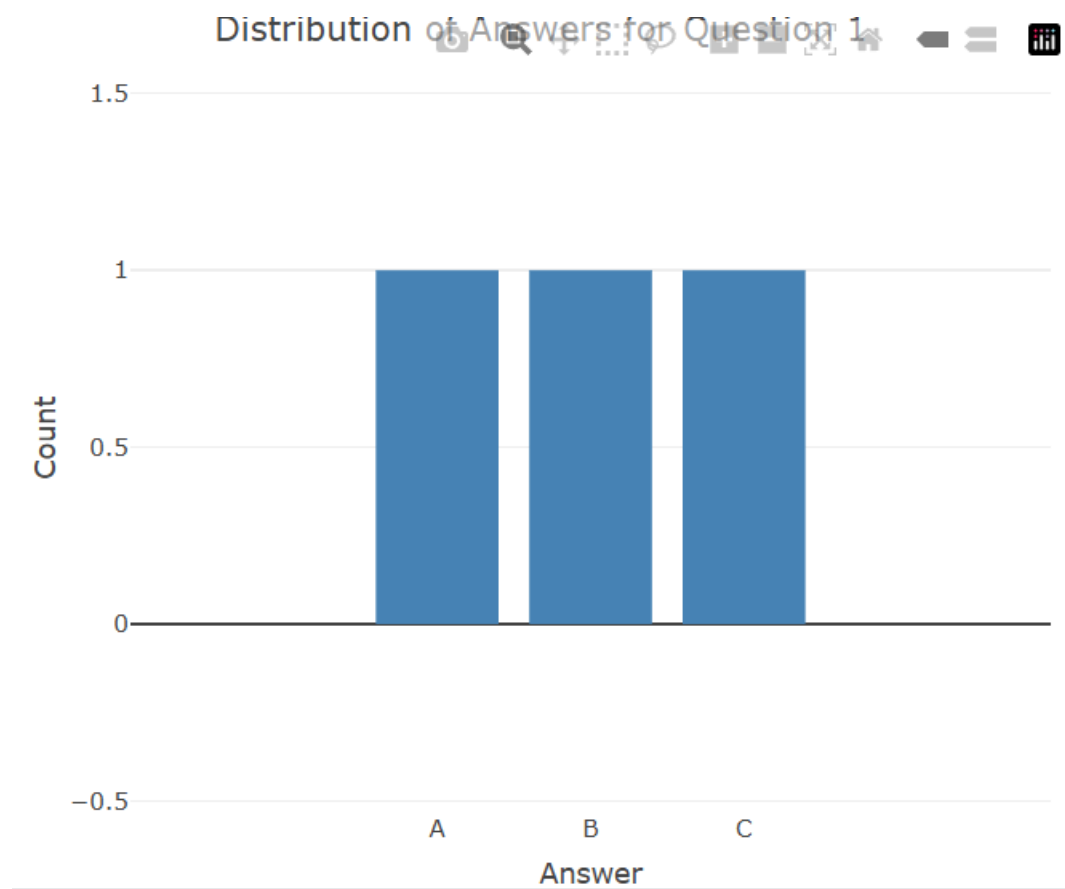
```
  title = "Distribution of Answers for Question 1",
```

```
xaxis = list(title = "Answer"),  
yaxis = list(title = "Count")  
)
```

```
# Display Bar Chart
```

```
bar_chart
```

output:



Set 15: Product Sales Analysis

- Create a grouped bar chart to visualize the sales of each product for the first quarter. Label the chart elements.

R code:

```
# Install packages if not installed
```

```
install.packages(c("ggplot2", "tidyr"))
```

```
# Load libraries
```

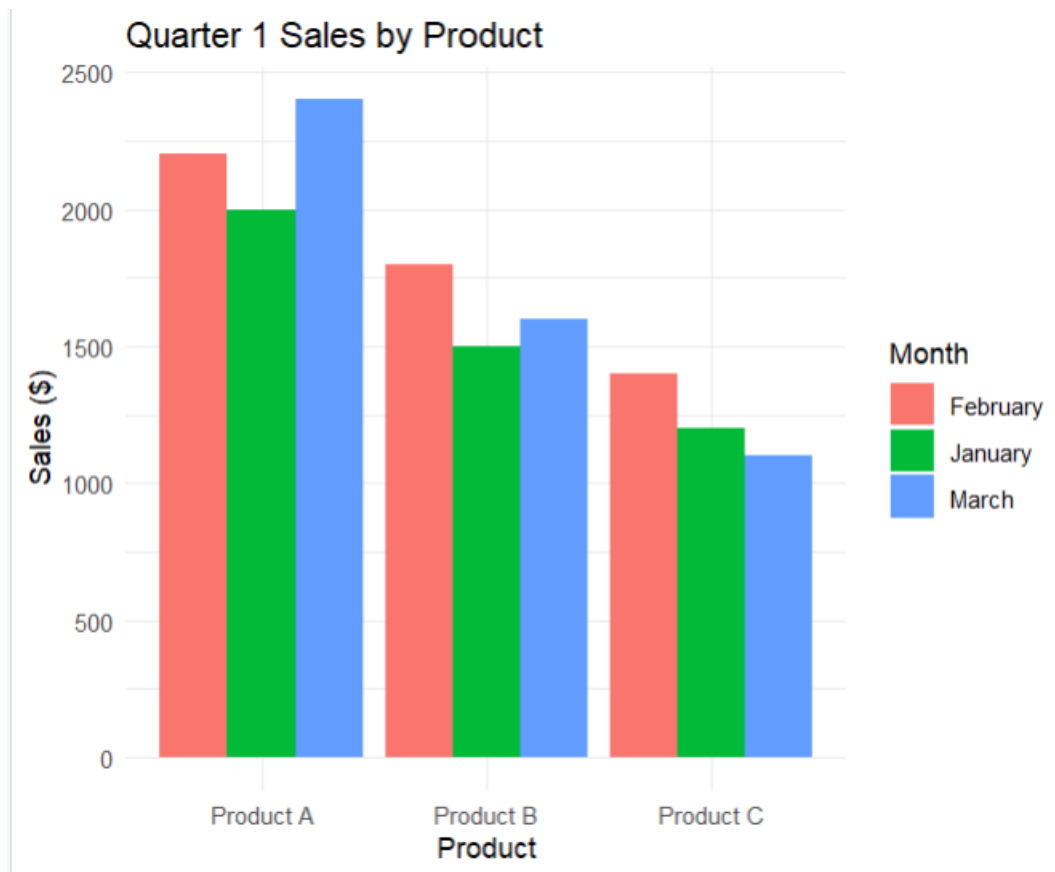
```
library(ggplot2)

library(tidyr)

# Convert data to long format
sales_long <- pivot_longer(
  sales_data,
  cols = January:March,
  names_to = "Month",
  values_to = "Sales"
)

# Grouped Bar Chart
ggplot(sales_long,
  aes(x = ProductName,
    y = Sales,
    fill = Month)) +
  geom_bar(stat = "identity",
    position = "dodge") +
  labs(
    title = "Quarter 1 Sales by Product",
    x = "Product",
    y = "Sales ($)",
    fill = "Month"
  ) +
  theme_minimal()

output:
```



- b) Generate a stacked area chart to represent the overall sales trend for all products over the three months.

R code:

```
# Install packages
```

```
install.packages(c("ggplot2", "tidyr"))
```

```
library(ggplot2)
```

```
library(tidyr)
```

```
# Convert data
```

```
sales_long <- pivot_longer(
```

```
  sales_data,
```

```
  cols = January:March,
```

```
  names_to = "Month",
```

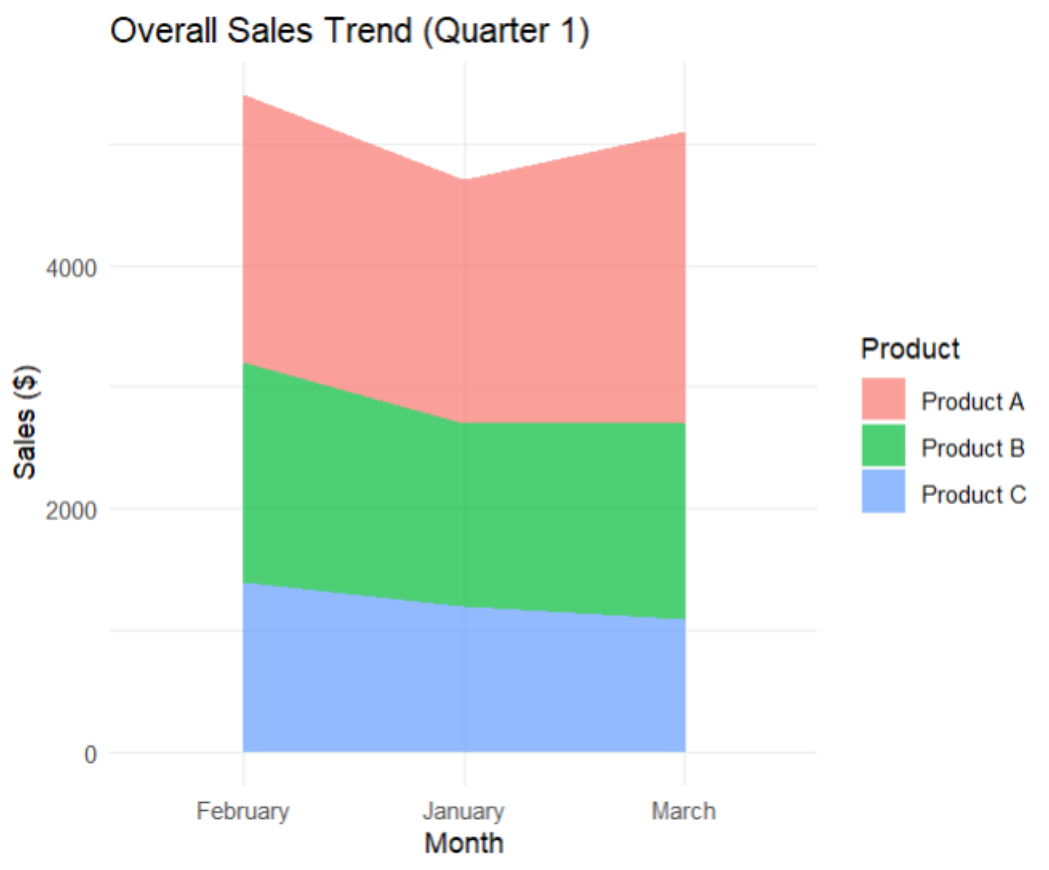
```
  values_to = "Sales"
```

)

Stacked Area Chart

```
ggplot(sales_long,
  aes(x = Month,
    y = Sales,
    fill = ProductName,
    group = ProductName)) +
  geom_area(alpha = 0.7) +
  labs(
    title = "Overall Sales Trend (Quarter 1)",
    x = "Month",
    y = "Sales ($)",
    fill = "Product"
  ) +
  theme_minimal()
```

output:



- c) Build a table to show the monthly sales data for each product. Label the table elements.

R code:

```
# Display dataset
```

```
print(sales_data)
```

```
# Optional formatted table
```

```
install.packages("gridExtra")
```

```
library(gridExtra)
```

```
grid.table(sales_data)
```

output:

	ProductID	ProductName	January	February	March
1	1	Product A	2000	2200	2400
2	2	Product B	1500	1800	1600
3	3	Product C	1200	1400	1100

- d) Develop a Tableau dashboard combining the grouped bar chart, stacked area chart, and the table for interactive exploration of sales data.

R code:

Install packages

```
install.packages(c("plotly", "gridExtra"))
```

Load libraries

```
library(plotly)
```

```
library(gridExtra)
```

Grouped Bar Chart

```
bar_chart <- plot_ly(
```

```
  sales_long,
```

```
  x = ~ProductName,
```

```
  y = ~Sales,
```

```
  color = ~Month,
```

```
  type = "bar"
```

```
) %>%
```

```
layout(
```

```
  barmode = "group",
```

```
  title = "Quarter 1 Sales by Product",
```

```
  xaxis = list(title = "Product"),
```

```
  yaxis = list(title = "Sales ($)")
```

```
)
```

```

# Stacked Area Chart
area_chart <- plot_ly(
  sales_long,
  x = ~Month,
  y = ~Sales,
  color = ~ProductName,
  type = "scatter",
  mode = "lines",
  stackgroup = "one"
) %>%
layout(
  title = "Overall Sales Trend",
  xaxis = list(title = "Month"),
  yaxis = list(title = "Sales ($)")
)

```

```

# Dashboard
subplot(
  bar_chart,
  area_chart,
  nrows = 1,
  margin = 0.08,
  titleX = TRUE,
  titleY = TRUE
)

```

Output:

